



Hypothesis

Origin of microbial life hypothesis: A gel cytoplasm lacking a bilayer membrane, with infrared radiation producing exclusion zone (EZ) water, hydrogen as an energy source and thermosynthesis for bioenergetics

J.T. Trevors^{a,*}, G.H. Pollack^{b,**}

^a Laboratory of Microbiology, School of Environmental Sciences, University of Guelph, 50 Stone Rd., East, Guelph, Ontario, Canada N1G 2W1

^b Department of Bioengineering, University of Washington, Box 355061, Seattle, WA 98195, USA

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ABSTRACT

The hypothesis is proposed that pre-biotic bacterial cell(s) and the first cells capable of growth/division did not require a cytoplasmic membrane. A gel-like microscopic structure less than a cubic micrometer may have had a dual role as both an ancient pre-cytoplasm and a boundary layer to the higher-entropy external environment. The gel pre-cytoplasm exposed to radiant energy, especially in the infrared (IR) region of the EM spectrum resulted in the production of an exclusion zone (EZ) with a charge differential (–100 to –200 mV) and boundary that may have been a possible location for the latter organization of the first cytoplasmic membrane. Pre-biotic cells and then-living cells may have used hydrogen as the universal energy source, and thermosynthesis in their bioenergetic processes. These components will be discussed as to how they are interconnected, and their hypothesized roles in the origin of life.

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1. Introduction

After decades of research on the origin of bacterial life on the Earth and possibly elsewhere [1], there is no agreed upon theory for the cellular origin of life. We present a hypothesis for some interconnected possibilities for the origin of cellular life. Cells can be considered soft matter; defined as materials composed of particles with dimensions from 1 nm to 1 µm and dispersed in a solvent. Soft matter such as cells also contains polymers, ions and salts, [2] on the Earth and/or possibly elsewhere. The recent observations of Hoover [3] provided evidence of microbial fossils similar to cyanobacteria in internal locations of freshly fractured carbonaceous meteorite samples. Although still debated, this discovery has profound implications for the plausible origin of life elsewhere in the universe and its delivery to the Earth billions of years ago by impact events.

These are exciting times for science and humanity. Possibly DNA and RNA sequences will be recovered from meteorite samples.

Whether life originated on the Earth or elsewhere, scientists are still confronted with understanding a plausible structure for the first cell(s) capable of regulated metabolism, growth and division. During bacterial cell growth and division, cells do not synthesize a completely new cytoplasmic membrane. Instead, an existing cytoplasmic membrane is augmented by the gradual addition of new, correct, and properly located membrane molecules. This means that over billion years of evolution on the Earth and/or elsewhere, the cytoplasmic membrane has never been completely resynthesized, but only altered during evolution and added to in sufficient amounts to enable growth and cell division. The role of the bilayer cytoplasmic membrane (CM) in present-day life forms is becoming better understood through research advances. However, in this article we hypothesize that a bilayer membrane was not essential for the organization of the first bacterial cells on the Earth or elsewhere.

Bacterial cytoplasmic membranes are mostly composed of amphipathic phospholipids and lesser amounts of glycolipids, and have been described in the excellent chapter by Saier [4]. Present-day bacterial cytoplasmic membranes are fluid bilayers [5–7] that also contain embedded and peripheral membrane proteins. However, the composition and structure of the first ancient bacterial membrane is still an enigma. An understanding of the cytoplasmic membrane is central to understanding a living cellular

* Corresponding author. Tel.: +1 519 824 4120x53367; fax: +1 519 837 0442.

** Corresponding author. Tel.: +1 206 685 1880; fax: +1 206 685 3300.

E-mail addresses: jtrevors@uoguelph.ca (J.T. Trevors), ghp@u.washington.edu (G.H. Pollack).

structure capable of enclosing cytoplasm, organic genetic material (DNA and RNA), ribosomes and all other necessary components in the first bacterial cell(s) and subsequent evolution. The origin of life was simpler and may not have required a bilayer membrane. A few, low molecular weight, interacting, non-limiting, components are more plausible for an origin of life hypothesis.

Decades of research have been undertaken to understand the cytoplasmic membrane in cell(s). A structurally complex, continuous, closed, fluid, semi-permeable, bilayer, cytoplasmic membrane of about 7 nm in width is a normal ubiquitous structure in present-day bacteria. Even in intact bacterial spores, the cytoplasmic membrane is still present [8]. Additional challenges in this knowledge quest include the microscopic location/environment where the first membrane became organized in the macroscopic universe (Earth and/or elsewhere), the correct genetic instructions for membrane synthesis and their origin at the same microscopic location, and subsequent alterations in membranes during billions of years of evolution. The first cell(s) would eventually have required a mechanism to replicate and partition the genetic material and cytoplasm between two offspring cells. These cells eventually synthesized the correct components for the cytoplasmic membrane, and added them to the boundary layer of a gel to organize a more complex fluid bilayer membrane.

In addition, the first cell(s) capable of growth/division may not have required a major cell-division protein such as the FtsZ protein. Based on the research of Lluch-Senar et al. [9], cell division in a minimal bacterium such as the *Mycoplasma genitalium* (ftsZ null mutant) is possible using the force generated by its motile gliding machinery and its cytoskeleton. This raises the question as to whether the first cell division(s) in bacteria were possible in the absence of a cytoplasmic membrane and cell division proteins, which can be plausibly hypothesized to be products of bacterial evolution after the first cells had been evolving for some time period.

In researching the origin of life, researchers are confronted with an immense number of interrelated difficulties. One significant difficulty/challenge is attempting to describe the microscopic physical environment and mechanisms that lead to the first, living, soft matter, single-celled, prokaryotic life. Coupled with this challenge, is the inability to develop a correct theory with supporting evidence that explains the origin of living microorganisms [10]. The law of parsimony approach or Occam's (Ockham's) razor approach is applicable here, as some researchers are seeking an explanation and/or mathematical relationship that makes minimal assumptions, allows a plausible hypothesis to be formulated and tested, and explains all aspects of the origin of life. This is an immense research challenge, but it may have important implications in both understanding the origin of life and designing the first single-celled artificial life form capable of metabolism, growth and division.

2. The first bacterial cell(s) did not have a cytoplasmic membrane

Viable bacterial cells are soft matter composed of polymers [11] with a fluid cytoplasm and membrane boundary. However, the first pre-cells may have lacked some or all of the genetic instructions and resultant biochemical or bio-machinery [12], and the capability to assemble a membrane, as we now know it. It has been hypothesized that the first membranes may have been generated from short-chain fatty acids in a pre-biotic environment. However, another plausible hypothesis is that the first viable bacterial cells capable of growth and division may have lacked a distinct bilayer membrane. Research over the past several decades has provided an immense amount of knowledge on the structural organization and function of the cytoplasmic membrane in bacteria and other

species, including many eukaryotes. Nevertheless, the origin of the first, fluid, bilayer, cytoplasmic membrane and any associated membrane proteins and functional membrane bioenergetics are still an enigma.

Mulkidjanian et al. [13] hypothesized the co-evolution of primordial membranes and membrane proteins. However, this does not solve the question of how an organizing membrane and proteins occurred at the same microscopic location. Nor does it solve the problem of what the precursors to the modern day membrane were composed of, and their origin. The composition and structure of the first cytoplasmic membrane(s) is debatable because of the paucity of knowledge on the subject matter. There are simply knowledge gaps on the microscopic locations of the origin of life [14,15], the exact components that organized in pre-biotic life, and the transition or tipping point (and time required) to living cell(s). The time duration is not as important as the capability to overcome the local, microscopic, higher-entropy (more random) environment.

It is a challenge to hypothesize, test and understand the origin of bacterial life on the Earth or elsewhere, without a membrane or boundary, and a means to obtain, store and use energy. However, a gel-type environment provides an internal environment as well as boundary layer that interfaces with the environment external to the gel. The gel is also sufficiently stable to allow molecular entrapment and organization of molecules and even concentrating molecules required for life. Diffusion of molecules is initially adequate until a more organized cell evolves with a bilayer membrane, an increasing genome with the correct genes needed, and a shorter cell cycle time that permitted faster cell division and therefore more rapid evolution.

3. Gel cytoplasm, IR radiation, ribosomes and thermosynthesis

It is hypothesized that the organization of formerly dispersed molecules in a gel-like structure, less than $1\ \mu\text{m}^3$ with short diffusion distances, negatively charged (-100 to -200 mV range) exclusion zone (EZ) chemistry and infrared radiant energy [16] may have been sufficient for the first living cell(s). The energy source that is especially effective and present is the sun's radiant energy in the $3.1\ \mu\text{m}$ wavelength [16]. It is noteworthy that this radiation wavelength is also included in the cosmic infrared background radiation, which is the radiation from the early universe after the Big Bang [17]. If this background IR radiation had a role in the organization and emergence of life, it is certainly abundant in the universe.

An illustration of the hypothesized processes described is presented in Fig. 1. It is hypothesized that EM radiation generates an exclusion zone (EZ) with a negative charge inside the gel and also outside, immediately adjacent to the gel boundary. The gel may have been attached to a mineral surface but this is not necessary. The gel with the EZ is the required structure. The charge differential in the -100 to -200 mV range at the EZ boundary may have been a plausible location for the latter assembly of a primitive cytoplasmic membrane. Present-day bacterial cytoplasmic membranes (e.g., *Escherichia coli*) also have a charge differential in the -85 to -154 mV range depending on the pH; with an average change of about -22 mV per pH unit [18]. This charge differential is in the same magnitude as that generated in the exclusion zone. This may be coincidence or it may have significance in establishing a charge boundary for eventual membrane organization. It is hypothesized that the initial processes in the organization of pre-biotic life were physical–chemical events with no organic, genetic instructions present. As organization proceeded, the organizing microscopic gel structure made the transition to a gel with a charged boundary layer to which phospholipids, proteins and a more rigid cell wall

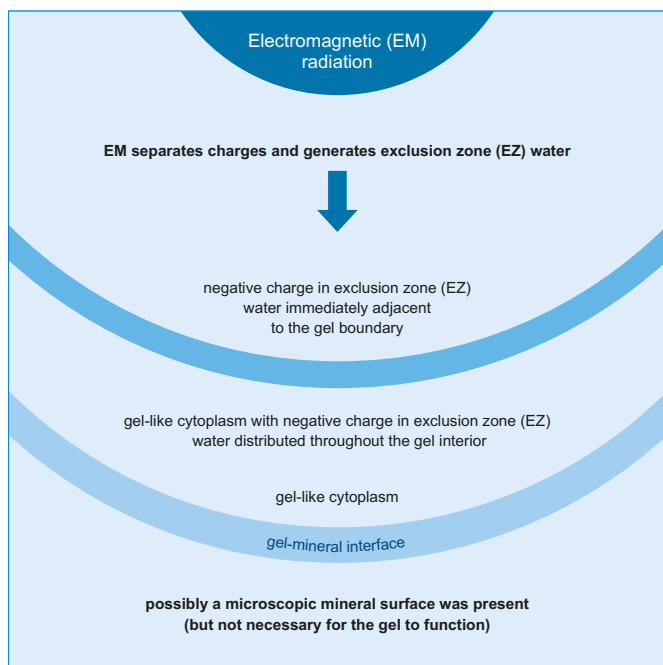


Fig. 1. Illustration of an hypothesized mechanism whereby EM radiation generates an exclusion zone (EZ) with a negative charge both inside and also outside, immediately adjacent to the gel boundary. The gel may have been attached to a mineral surface but this was not necessary. The gel with the EZ is the required structure. The charge differential in the -100 to -200 mV range at the EZ boundary may have been a plausible location for beginning of the assembly of an eventual primitive cytoplasmic membrane. Present-day bacterial cytoplasmic membranes (e.g., *E. coli*) also have a charge differential in the -85 to -154 mV range depending on the pH; with an average change of about -22 mV per pH unit (18). As organization proceeded, the organizing microscopic gel structure made the transition to a gel with a charged boundary layer to which phospholipids, proteins and a more rigid cell wall were attached once organic, genetic instructions were available and enclosed in the gel-like cytoplasm.

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The possibility that a bilayer, cytoplasmic membrane was not present in the first living bacteria also fits with the concept of a minimal bacterial cell with a small genome [19]. The first minimal cell with minimal dimensions and a minimal genome may have been assisted in its functions by IR radiant energy. The more complex membrane that we research today is a product of evolution, once bacterial cells were present that could metabolize and divide; and the correct genetic instructions were present for assembly of a more advanced membrane. However, what was needed for the origin of the first bacterial life was much simpler, and composed of molecules available in the required and always available concentrations (i.e., not limiting) and subject to physical–chemical events present in the universe. The specific proteins that became part of the cytoplasmic membrane could have been added after the cell was capable of growth/division and was subjected to natural selection and Darwinian evolution. The outer membrane (OM) of Gram-negative bacteria can also be hypothesized to be a latter addition, once evolution had commenced.

EM radiation especially in the IR range may have another useful function in bacterial cells, assisting in the movement of cytoplasm. Cytoplasmic streaming apparently has not been observed in bacteria. mRNA locates abundant ribosomes by diffusing through the crowded but spatially organized cytoplasm [20]. The question can be posed; did or does radiant energy make EZ water, or the proton-rich water contiguous with the EZ, move in bacteria and

does this facilitate the movement of mRNA and other molecules? This might not be the same as true cytoplasmic streaming but it may be radiant energy producing movement of cytoplasm in a microscopic cell.

The mechanism by which mRNA transcripts contact ribosomes in bacteria has been hypothesized as simple diffusion in a spatially crowded cell. For example, in *E. coli*, the number of ribosomes per cell has been estimated to range from 6800 to 72,000 depending on the growth rate [20]. The shorter the cell doubling time, the more ribosomes are present. The mRNA transcripts may move by diffusion in the cytoplasm, but the large 70S bacterial ribosomes likely do not diffuse, but can occupy from 6 to 15% of the total cell volume [20]; see also Cyber-cell data base, <http://redpoll.pharmacy.ualberta.ca/CCDB/> for other values. The mRNA transcripts remain close to the transcription site and must quickly contact ribosomes during rapid cell growth and division. However, if radiant energy moves cytoplasm, then the transcripts are moved about and easily come into contact with the thousands of ribosomes – a type of molecular collision event suited to a structured gel-like cytoplasm. The chance that transcripts will gently collide with thousands of ribosomes is very good indeed. Ribosomes may even gently collide with other ribosomes, with no damage incurred.

The hypothesis is put forth that cytoplasm can be moved by radiant EM radiation. This needs to be tested experimentally and it is only a hypothesis at this time for development, thought experiments and discussion purposes. It is recognized that not all bacterial cells are exposed to EM radiation from the sun. However, cells produce their own radiant energy as well as receive radiant energy from geothermal sources and the cosmic background IR is present in the universe.

At temperatures above absolute zero, atoms in molecules are in continuous motion. If the frequency of the vibration is equal to the frequency of an IR source, the molecule absorbs the radiation. The two major molecular vibrations are bond stretching and bending. As the IR radiation is absorbed, the associated energy is converted into these vibrations. For example, the OH combination from water has been assigned at 5200 cm^{-1} ($1.92\text{ }\mu\text{m}$) [21]. Bond stretching of pre-biotic molecules (stretching requires more energy than bending) and bending caused by IR radiation in an organizing primitive gel may have been plausible. The role played in EZ water and the organization of pre-biotic cells and then evolving cells still remains to be determined. However, Antonyuk [22] proposed that broad band light transports electrons in random media in a direction opposite to the direction of the electrical force and may be responsible for self-organization events, which are required for the origin of life. This information may lend some support the IR driven organization of an EZ in a microscopic gel environment.

Some bacterial mRNAs localize near their synthesis site on the genome as well as move from the genome to other cellular sites for subsequent translation [23]. It would be valuable to know if the targeting of a site in the cell is a non-random mechanism of the mRNA [23]. Some mRNAs seem to contain information that determines their non-random movement to localize at sites in the cell where the translated proteins are necessary? This can be hypothesized to be site targeting and/or may be even a streaming process. In addition, ribosomes associated with the cytoplasmic membrane in *E. coli* that are involved in the translation of membrane-bound proteins have been reported [24]. This type of sophisticated ribosome localization can be hypothesized to have been absent in pre-biotic cells and the first living bacterial cells. In the ancient cells, processes like diffusion, Brownian motion, and, as proposed in this article, EZs and radiant driven cytoplasmic movement, may have been central to the organization of the first cell(s). What happened first for the origin of life was much simpler until Darwinian evolution proceeded.

A gel cytoplasm with an initial distinct boundary can also allow hydrogen and some other gases and molecules to enter *via* diffusion. This entry may be significant, as Morita [25] hypothesized that hydrogen was ubiquitous on the early Earth and may have been a universal energy source. Present-day bacterial cells can utilize hydrogen in some biochemical reactions and some coliform bacteria still utilize hydrogen as an energy source, but not for growth. Growth requirements are now satisfied by other substrates. A simple organizing gel, permeable to hydrogen by diffusion, may have been the precursor to the origin of the first cell(s), which plausibly used hydrogen as a universal energy source.

Another plausible mechanism operating as a source of energetics in the origin of life may have been thermosynthesis [26]. A thermal gradient provided the mechanism for a microscopic heat engine that performed condensation reactions and was postulated as a precursor of the ATP synthase. Thermosynthesis also synchronizes the breakage of bonds and rebonding [26]. In thermosynthesis, the contents of inanimate pre-cell(s) are thermally cycled and eventually produce polymer end products. However, the model presumes the presence of both protein and RNA precursors [26]. In the pre-cytoplasm and then cytoplasm gel, without a cytoplasmic membrane, the generation of an EZ assisted by IR radiation, with hydrogen as the first energy source may have present. Thermosynthesis (energy gains) would be present prior to microbial fermentation, respiration and photosynthesis [26], which all came later in the evolutionary history of life.

In thermocycling, which is also present on the Earth as the temperature increases and decreases, IR radiation is emitted. This would be useful in generating EZs and breaking some bonds. Cooling would decrease the radiation and bonds could reform. It was not only the temperature cycling, but also the accompanying IR that had a role in the first physical–chemical organization of primordial life. It is possible that IR radiation not only warmed the Earth, but also had a role in structuring water and producing an EZ next to a microscopic gel with a negative charge separation, that was involved in stretching and bending bonds and assisting in the movement of molecules in the gel cytoplasm. Physical–chemical mechanisms such as heating and cooling were present before the presence of protein catalysts and organic genetic instructions in RNA and DNA.

Thermosynthesis and temperature cycling were not efficient compared to enzyme reactions. They were however, sufficient when occurring in a gel with a non-membrane boundary, EZ water formation assisted by IR radiation and available hydrogen as an ubiquitous energy source. In present-day research, thermocycling is commonly used to denature DNA at higher temperatures, followed by a decreased temperature for reannealing and then extension reactions such as in the polymerase chain reaction (PCR). The role of thermocycling in a microscopic gel along with the other mechanisms discussed in this article are still in the hypothesis stage. However, they may be plausible, interconnected, physical–chemical mechanisms in a gel, which, when combined together assisted in the organization of pre-biotic life.

4. Systems biology and artificial living cells

It is possible that a better understanding of radiation, gels, hydrogen, water, thermosynthesis, thermocycling, charges, genomes, pre-membrane boundaries, exclusion zones (EZ) and boundaries, will assist in generating artificial life using a systems biology approach. Systems biology is not a new science [19]. The engineering of a synthetic bacterial cell (or other type of cell) capable of growth and division may be possible. This possibility has been addressed in the excellent article, “Building a blueprint for life” by Henry et al. [27]. Initially, it is plausible that simpler experiments using gels, surfaces,

and IR radiation will provide useful clues for bioengineering life forms initially stable and capable of metabolism. Controlled cell division will require significant additional research efforts. For a bacterial cell this will require a complete circular, core or minimal synthetic genome where the exact correct function of every essential gene and translated protein is understood and the cytoplasm is suitable (e.g., amount, volume, pH, sufficient number of ribosomes, no limiting ingredients) for cell division.

The synthesis of the first, synthetic, bilayer, cytoplasmic membrane with the correct fluidity and the capability to form a mid-cell septum and partition the cytoplasm and replicate the genome correctly will be a biological engineering breakthrough. Engineering such a synthetic cell may require an integrated or systems approach. Genome-driven cell engineering with stable cytoplasmic membrane engineering that allows cell division and the correct amount of cytoplasm, with a sufficient number of functional ribosomes for translation will be a challenge. The first artificial cells, if produced, may also have extended cell-division cycles.

Future transformative research will provide the data sets, successes and failures. Knowing that the laws of thermodynamics are the most fundamental of all laws, apply to the entire expanding universe, and govern all life forms [28,29], it is also logical to suggest that a thermodynamic approach needs to be considered as living cells are open thermodynamic systems with semi-permeable membranes far from equilibrium with the higher-entropy external surrounding environment.

5. Perspective, summary, and the future

It is hypothesized that a gel was sufficient for the origin of life (organized soft matter) in the absence of a membrane. The generation of EZ water adjacent to and also inside the gel, the hypothesized role of IR in EZ generation, hydrogen as the universal energy source and environmental thermal cycling have been discussed. An origin of life hypothesis is more plausible when the required component molecules and compounds are few and precursors are available.

This article has discussed the possibility that the first living cell(s) may have not contained a cytoplasmic membrane as known today. A simpler but still functional gel system may have been the precursor of the modern day cytoplasm and the cytoplasmic membrane. The postulated features necessary for the beginning of an ancient cell structure have been summarized in Table 1. For billion of years in the evolutionary history of life as we understand it today, the original cytoplasmic membrane has undergone countless accretion events in countless cells in diverse species on the Earth, and even possibly elsewhere in the universe. Accompanying these membrane accretion events have been the

Table 1

Possible characteristics of a primitive microscopic gel with the dual function of cytoplasm and a boundary (not a continuous membrane) to the external environment.

- gel-like structure
- microscopic in size (e.g., less than μm^3)
- possibly attached to mineral surfaces
- IR driven water and molecule movement in the gel
- diffusion processes occurring in and out of the gel
- short diffusion distances (nm to μm)
- exclusion zone (EZ) water chemistry was present and acted to produce altered pH gradients which are necessary for advanced membrane function, but may have been sufficient in the absence of cytoplasmic membranes and charge effects at boundaries
- abundant IR radiation as a radiant energy source
- hydrogen as a universal energy source and thermocycling bioenergetics

ancestors of the first cytoplasm and the organic genetic code. The cell as the basic unit of life, capable of regulated growth and division with its now known major components, the genetic code, the cytoplasm and the cytoplasmic membrane (the three C's of life; code, cytoplasm and cytoplasmic membrane) have overcome local microscopic and macroscopic entropy for billions of years. The origin of life and subsequent evolution on the Earth and elsewhere are truly amazing events, and life is still best understood and researched under the paradigm of evolution. The formulation of an origin of life theory with ample supporting evidence is the subject matter of many researchers. The correct answer(s) to the origin of life mechanisms is not obvious. Evolution was not obvious as well for many centuries. In the initial absence of the correct genetic instructions, proteins catalysts, bilayer membranes and advanced cell-division mechanisms, it is logical/rational to hypothesize cohesive gel structures, warming and cooling cycles, attempts to understand the role of EM radiation and a possible universal energy source such as hydrogen, which would have been abundant.

Some researchers are also thinking about and attempting the engineering of artificial living cells. Possibly gels will have a significant role in understanding both the origin of life, evolution and functional artificial cell engineering. If and when synthetic life is engineered, it may provide profound insights as to the origin of life, and/or there may be numerous plausible ways that life could have originated and new hypotheses will be tested. We invite any collaborations and ideas to solve the enigma of the origin of life.

Conflicts of interest

No conflicts of interest.

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